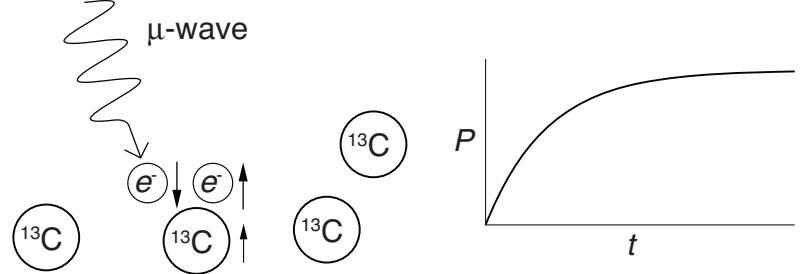
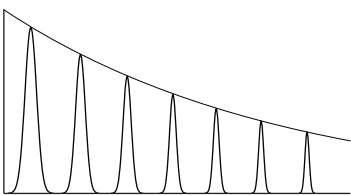


Rapid, high polarization

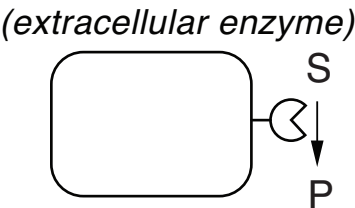
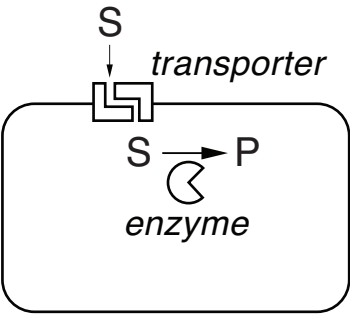


Slow *in vivo*
 T_1 & T_2 relaxation

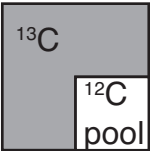
- ^{13}C at mobile sites
without nearby ^1H



Rapid uptake in
target tissue / cells

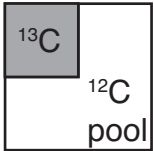


High fractional
enrichment
(tolerance test-like,
highest metabolite
production)



OR

Low fractional
enrichment
(tracer-like, minimize
metabolic perturbation)



Fast conversion to
long-lived
metabolite signals

- Δ chemical shift > 2 ppm
- Effectively 1-way reaction
(irreversible, or low label
enrichment in product)

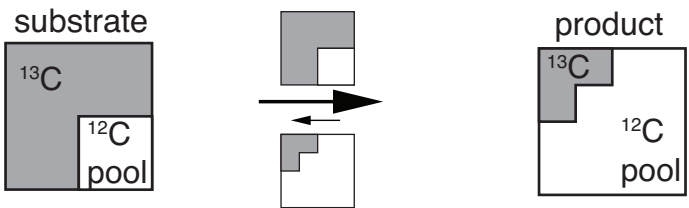
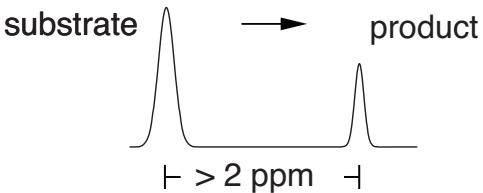


Figure 6.8 Characteristics of good HP ^{13}C agents.